

Jumping Spherical

Problem Description

A ball is released at rest and falls onto the ground. There is air resistance acting on the ball and its impact with the ground causes it to experience a stopping force in the direction opposite to its motion.

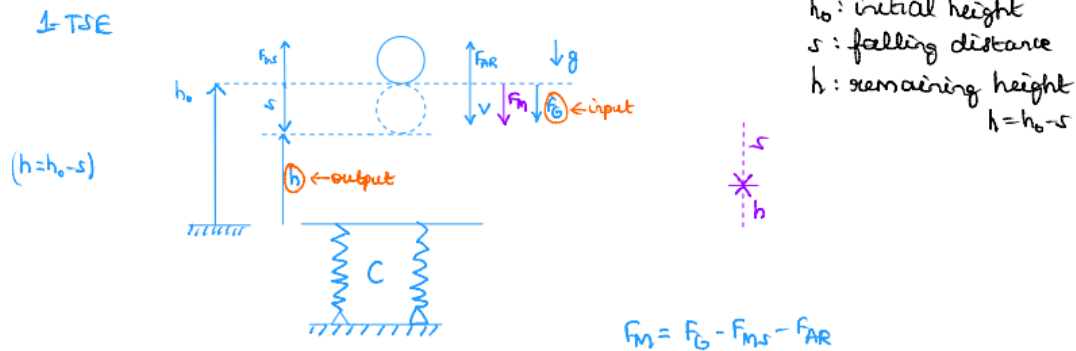
The force F_{AR} represents the air resistance force and the force F_c represents the force of the ground on the ball as it bounces off it.

The equations used are as follows:

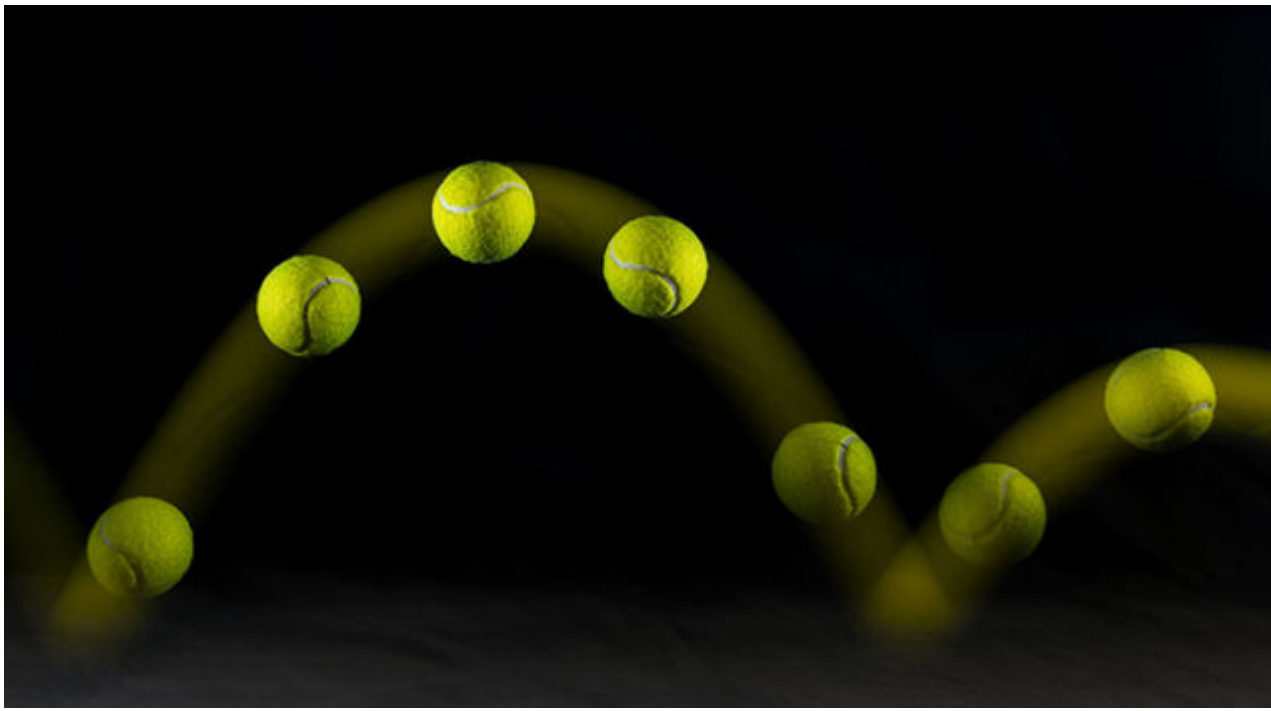
$$F_m = F_G - F_{AR} - F_C$$

$$F_{AR} = k_{AR} v_m^2 \text{sign}(v_m)$$

$$F_C = ch = c(h_0 - s)$$



Active force of spring pad when $h \leq 0$
 $[h = h_0 - s]$



Declaring Variables

```
% h represents the height of the ball above the ground:  $h = h_0 - s$ 
% where s is the displacement of the ball from its dropoff position
simtime = 3;
g = 9.81;           % acceleration due to gravity in N/kg
m = 0.1;            % mass in kg
h_0 = 1;            % drop off height in m
c = 10000;          % ground stiffness constant in N/m
                    % cannot be too low, else ball "goes below ground"
k_AR = 0.25;        % air resistance coefficient in  $N/(ms^{-2})$ 
```

Simulation and Behaviour Plot

```
sim('Jumping_Spherical_System.slx')
figure
subplot(4, 1, 1)
plot(h.time, h.data)
grid on
xlabel('t/s')
ylabel('h/m')

subplot(4, 1, 2)
plot(v_m.time, v_m.data)
grid on
xlabel('t/s')
ylabel('v_m/m')

subplot(4, 1, 3)
plot(F_AR.time, F_AR.data)
```

```

grid on
xlabel('t/s')
ylabel('F_A_R/N')

subplot(4, 1, 4)
plot(F_c.time, F_c.data)
grid on
xlabel('t/s')
ylabel('F_c/N')

```

